

reServer Industrial Reference Guide



Fanless compact AI-enabled NVR server

Wider temperature support -20 ~ 60°C with 0.7m/s airflow



Multi-stream processing

5* GbE RJ45(4 for 802.3af PSE); support multiple 4K streams with real-time processing complex tasks



Expandable storage

2 Drive Bays to support 2.5" SATA HDD/SSD; 1* M.2 2280 M key socket for NVMe SSD



Design for industrial interfaces

1* COM(RS232/422/485); 4* DI/DO; 1* CAN; 4* USB3.1; 1* TPM2.0 (Module optional)



Hybrid connectivity

Support 5G/4G/LTE/LoRaWAN® (Module optional) with 1* Nano SIM card slot

Introduction

reServer Industrial-series offers fanless compact AI-enabled NVR server built with NVIDIA advanced AI embedded systems: Nvidia Jetson Orin™ NX/Orin™ Nano delivering AI performance from 20 TOPS to 100 TOPS. With pre-installed OS system, PoE interfaces, rich extension modules, industrial peripherals, Hybrid network connectivity, thermal management, expandable storage combined with decades of Seeed's hardware expertise, reServer Industrial Jetson is ready to help you accelerate and scale the next-gen AI products emerging diverse AI scenarios.

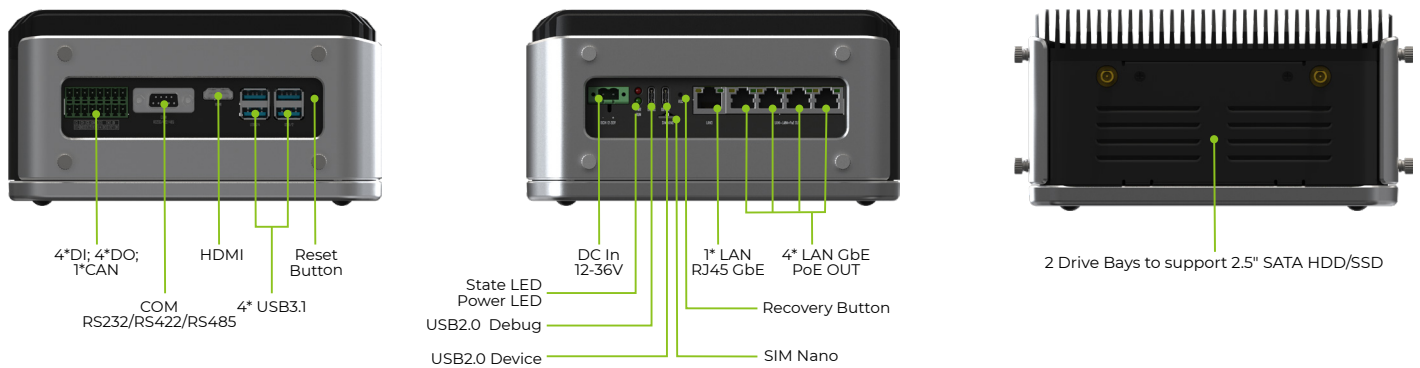
Packing List

- 1* reServer Industrial
- 1* DIN Rail Bracket
- 10* Bracket Screw
- 1* 16-Pin Terminal Block for DIO
- 1* 24V/5A Power Adapter(without power cord)
- 1* 2-Pin Terminal Block Power Connector

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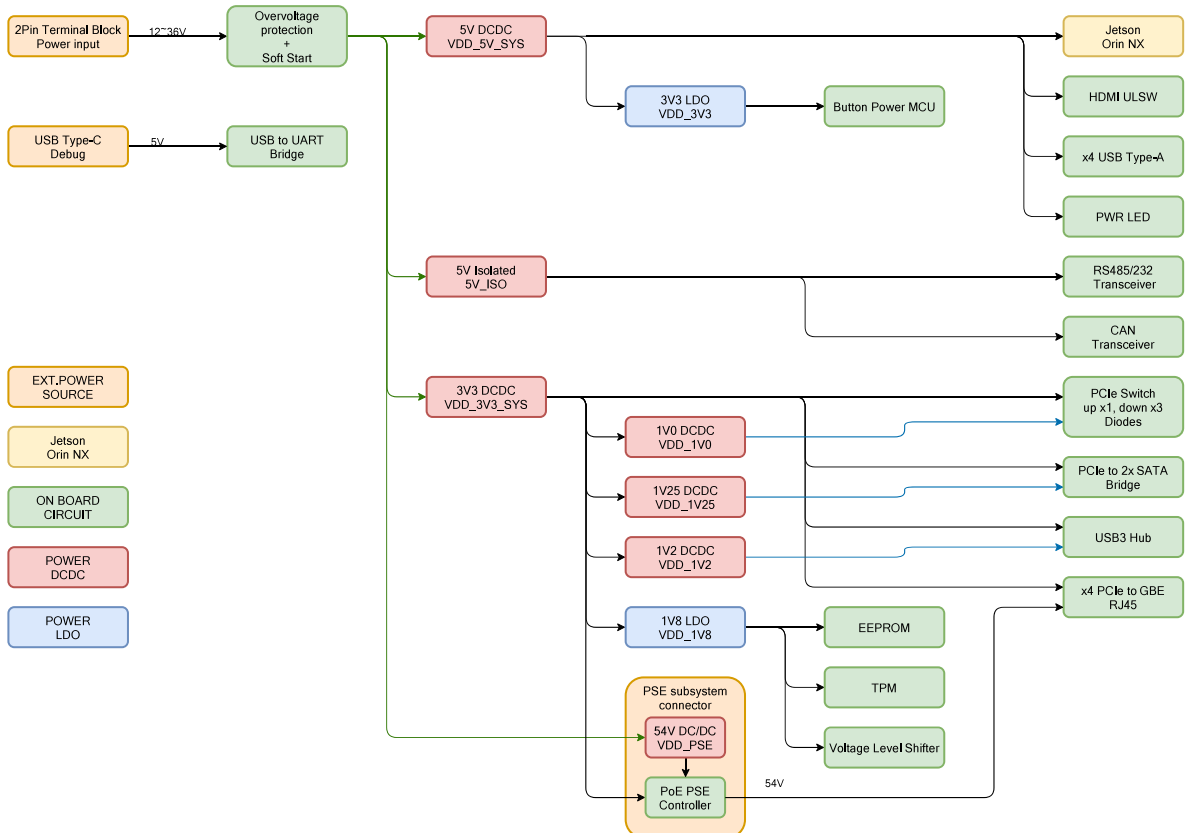
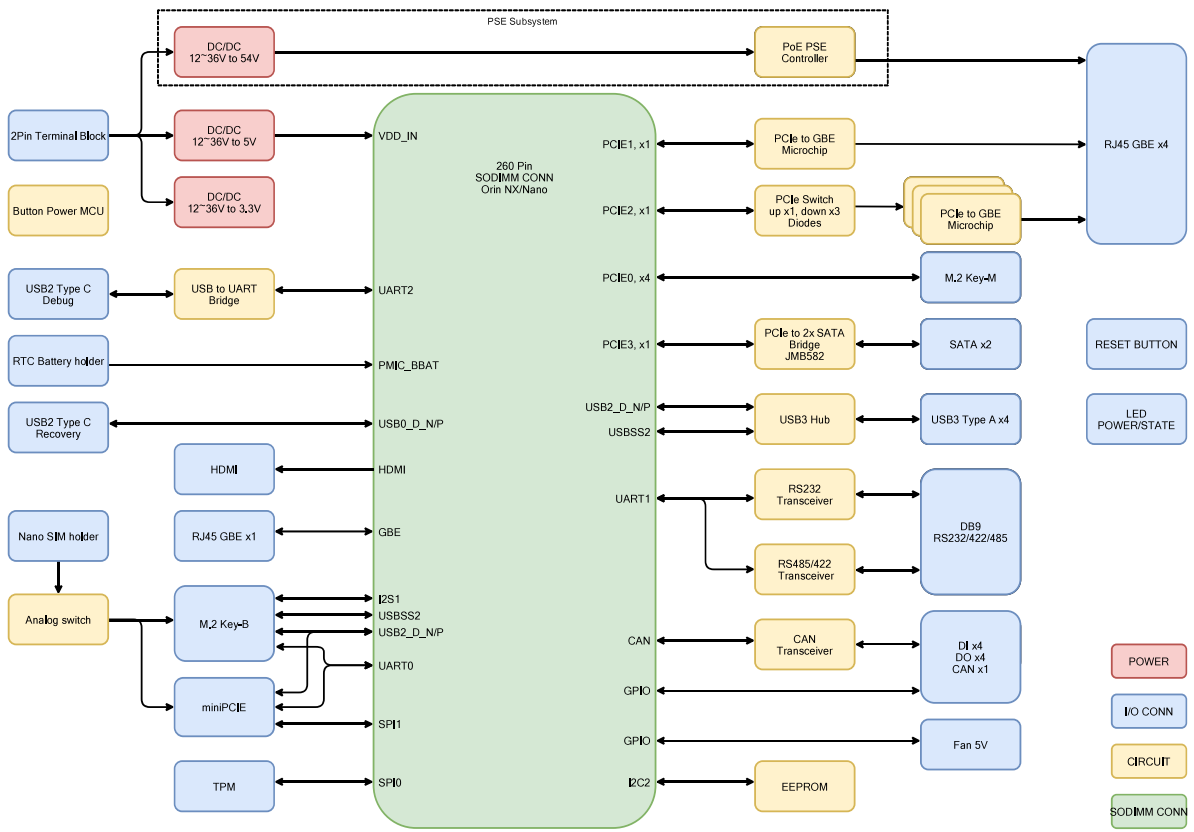
Hardware Overview



Specifications

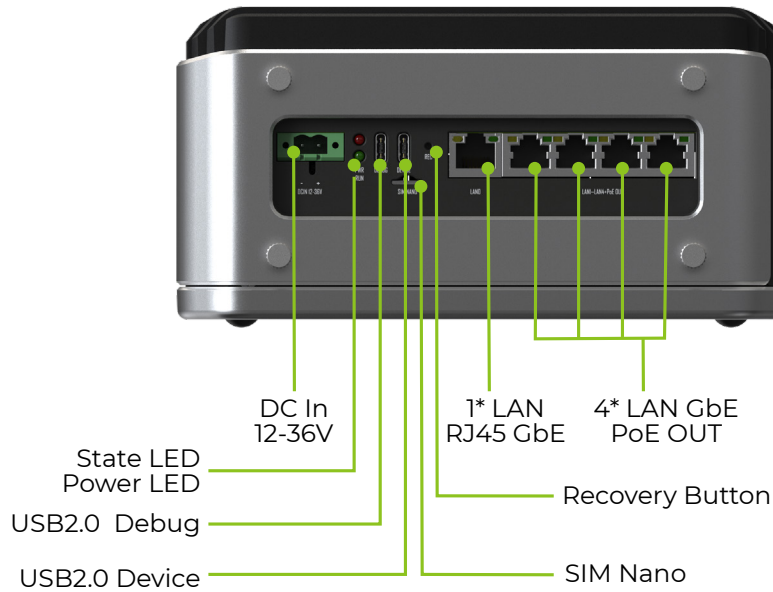
Product Name		reServer Industrial J4012	reServer Industrial J4011	reServer Industrial J3011	reServer Industrial J3010
NVIDIA Jetson Module		Orin NX 16GB	Orin NX 8GB	Orin Nano 8GB	Orin Nano 4GB
SKU		114110247	114110248	114110249	114110250
Processor System	AI Performance	100 TOPS	70 TOPS	40 TOPS	20 TOPS
	GPU	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores			512-core NVIDIA Ampere architecture GPU with 16 Tensor Cores
	CPU	8-core Arm® Cortex®-A78AE v8.2 64-bit CPU; 2MB L2 + 4MB L3	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3		
	Memory	16GB 128-bit LPDDR5 102.4GB/s	8GB 128-bit LPDDR5 102.4GB/s	8GB 128-bit LPDDR5 68 GB/s	4GB 64-bit LPDDR5 34 GB/s
	Video Encode	Standards supported: H.265 (HEVC), H.264, AV1 1*4K60 (H.265) 3*4K30 (H.265) 6*1080p60 (H.265) 12*1080p30 (H.265)			1080p30 supported by 1-2 CPU cores
	Video Decode	Standards supported: H.265 (HEVC), H.264, VP9, AV1 1*8K30 (H.265) 2*4K60 (H.265) 4*4K30 (H.265) 9*1080p60 (H.265) 18*1080p30 (H.265)			Standards supported: H.265 (HEVC), H.264, VP9, AV1 1*4K60 (H.265) 2*4K30 (H.265) 5*1080p60 (H.265) 11*1080p30 (H.265)
Storage	eMMC	-			
	Expansion	M.2 Key M (2280) PCIe Gen4.0 SSD (M.2 NVMe SSD 128G included)			
I/O	Ethernet	1* LAN0 RJ45 GbE (10/100/1000Mbps)			
	USB	4* USB3.1, 1* USB2.0 Type C (Device mode), 1* USB2.0 Type C For Debug UART & RP2040			
	DI/DO	4*DI, 4*DO, 3*GND_DI, 2*GND_DO, 1*GND_ISO, 1*CAN			
	COM	1* DB9 (RS232/RS422/RS485)			
	Display	1*HDMI 2.1 Type A 7680x4320 at 30 Hz			1*HDMI 1.4 Type A 3840x2160 at 30 Hz
	SATA	2 Drive Bays to support 2.5" SATA HDD/SSD (SATA III 6.0Gbps)			
	SIM	1* Nano SIM card slot			
	Button	Reset Button, Recovery Button			
	Mini PCIe	Mini PCIe for LoRaWAN®/4G/5G Series Wireless (Module optional)			
Expansion	M.2 Key B	M.2 Key B (3042/3052) support 4G/5G (Module optional)			
	Fan	Fanless, passive heatsink; 1*Fan connectors(5V PWM)			
	TPM	1* TPM 2.0 connector (Module optional)			
	RTC	1* RTC socket (CIR1220 included), 1* RTC 2-pin			
Power	Power Supply	DC 12V-36V Terminal block 2 pin			
	Power Adapter	24V/5A Power Adapter (Without power cord)			
Mechanical	Dimensions (W x D x H)	194.33mm*187mm*95.5mm			
	Weight	2.8kg			
	Installation	Desk, DIN rail, VESA			
Environment	Operating Temperature	-20 ~ 60°C with 0.7m/s			
	Operating Humidity	95% @ 40 °C (Non-condensing)			
	Storage temperature	-40 ~ 85°C			
	Storage humidity	60°C@ 95% RH (Non-condensing)			
	Vibration	3 Grms @ 5 ~ 500 Hz, random, 1 hr/axis			
OS	Shock	50G peak acceleration (11 msec. duration, eMMC, microSD, or mSATA)			
		Pre-installed Jetpack 5.1.1 (above) (provide Linux OS with board support package)			
Certification		FCC, CE, RoHS, UKCA, KC			
Warranty		2 Years			

Block Diagram



Pin Definitions

Front View



DC in - J17

12V-36V, 10A current limit.

After plugging in the power supply, the POWER LED lights up.

LED

LED Name	Descriptions	Status
RUN	Green, system operation indicator	Always on or blink after the system is running
POWER	Red, 3.3V Power Indicator	Always on after power on

USB 2.0 DEVICE - J18

Pin #	Schematic Signal	Usage/Description	Type/Dir Default
A4/B9	Type C_UART_5V	VBUS Supply	Power
A9/B4			
A5	CC1_2	–	–
B5	CC2_2	–	–
A7	Type C_USB_DN	USB 2.0 #From MCU RP2040	Bidir
B7			
A6	Type C_USB_DP		
B6			
A1/B12	–	Ground	Ground
A12/B1	–		Ground

Note:

1. Print serial port information for debugging: Pin3 and Pin4 of J8 without jumper caps, at this time USB_MUX_SEL=1, Type C connects to the USB signal of the core board, it is USB to UART mode, at this time Type C prints the serial debug information of the core board.
2. Burning firmware to the RP2040: Pin3 and Pin4 of J8 with jumper cap, at this time USB_MUX_SEL=0, Type C connects to the USB signal of RP2040, which is the burn-in interface of RP2040, set RP2040 into burn-in mode, which can update the firmware of RP2040 (update method: after connecting Pin5 Pin6 of J8 with jumper cap, then connect Pin7 Pin8 to reset RP2040 and make it enter into burn-in mode).

USB 2.0 DEBUG - J19

Pin #	Module Pin Name	Module Pin #	Schematic Signal	Usage/Description	Type/Dir Default	
A4/B9	GPIO00	87	USB0_VBUS_DET*	VBUS Supply	Power	
A9/B4	(USB_VBUS_EN0)					
A5	–	–	CC1_1	–	–	
B5	–	–	CC2_1	–	–	
A7	USB0_D_N	109	USB0_AP_N	USB 2.0 #0 Data	Bidir	
B7						
A6	USB0_D_P	111	USB0_AP_P			
B6						
A8	–	–	–	–	–	
B8	–	–	–	–	–	
A1/B12	–	–	–	Ground	Ground	
A12/B1	–	–	–		Ground	

Note:

- 1.The computer can recognize the board information: USB 2.0 Device interface on the core board, which can use Type C cable for data communication.
- 2.Burning image to Jetson: While the board is powered up, press the RECOVERY button SW2 to enter the recovery mode, which allows you to burn in the system of the core board.

Nano SIM card slot - J6

PIN	Schematic Signal	PIN	Schematic Signal
C1	SIM_CLK	C2	SIM_DIO
C3	SIM_RST	C4	NC
C5	VDD_SIM	C6	GND
C7	SIM_DETECT	G1,G2,G3,G	GND_SHD

Note:

1. Only one of M.2 Key B and Mini PCIe can be used at a time.
2. The insertion status of SIM card can be detected by the signal SIM_DETECT_IV8, which is high when SIM card is inserted and low when it is pulled out.

RECOVERY Button - SW2

Recovery button: FORCE_RECOVERY, press the RECOVERY button SW2 to enter the recovery mode, which allows you to flash the system of the core board.

GbE connector

LAN LED connection speed color indicators:

	Active & Link (Left LED)	Speed (Right LED)
10 Mbps	Yellow (Blinking)	N/A
100 Mbps	Yellow (Blinking)	N/A
1000 Mbps	Yellow (Blinking)	Green

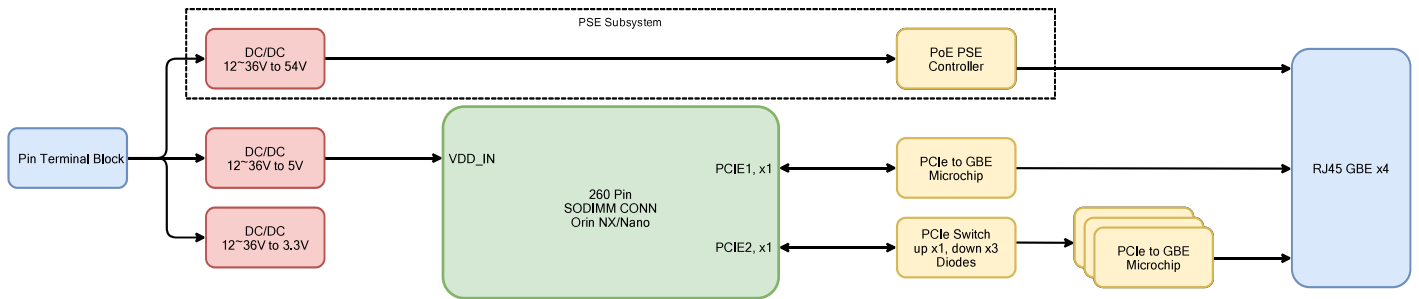
LAN0 - J14

Pin #	Module Pin Name	Module Pin #	Schematic	Usage/Description	Type/Dir
1	–	–	MCT	MCT	–
2	GBE_MDIO_P	186	GBE_MDIO_P	Gigabit Ethernet MDI 0+	Bidir
3	GBE_MDIO_N	184	GBE_MDIO_N	Gigabit Ethernet MDI 0–	Bidir
4	GBE_MDII_P	192	GBE_MDII_P	Gigabit Ethernet MDI 1+	Bidir
5	GBE_MDII_N	190	GBE_MDII_N	Gigabit Ethernet MDI 1–	Bidir
6	GBE_MDII2_P	198	GBE_MDII2_P	Gigabit Ethernet MDI 2+	Bidir
7	GBE_MDII2_N	196	GBE_MDII2_N	Gigabit Ethernet MDI 2–	Bidir
8	GBE_MDII3_P	204	GBE_MDII3_P	Gigabit Ethernet MDI 3+	Bidir
9	GBE_MDII3_N	202	GBE_MDII3_N	Gigabit Ethernet MDI 3–	Bidir
10	–	–	–	NC	
11	–	–	VDD_3V3_SYS	Main 3.3V supply	Power
12	GBE_LED_LINK	188	GREEN_LED	Green LED Cathode. On for 1000Mbps link. Off for 10/100Mbps.	Output
13	GBE_LED_ACT	194	YELLOW_LED	Yellow LED Cathode. On indicates activity.	Output
14	–	–	VDD_3V3_SYS	Main 3.3V supply	Power
15	–	–		Shield Ground	Ground
16					

Note:

1. Native GbE from NVIDIA® Jetson™ modules
2. Gigabit Ethernet (10/100/1000M)

4* LAN GbE PoE OUT - J10



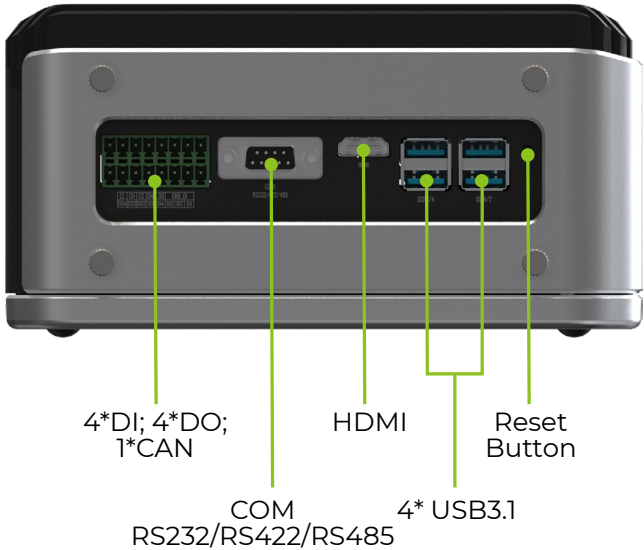
Pin #	Schematic Signal	Usage/Description	Type/Dir
LAN 1 - J10-1			
1	MXA_P	Gigabit Ethernet MDI 4+	Bidir
2	MXA_N	Gigabit Ethernet MDI 4-	Bidir
3	MXB_P	Gigabit Ethernet MDI 3+	Bidir
4	MXB_N	Gigabit Ethernet MDI 3-	Bidir
5	MXC_P	Gigabit Ethernet MDI 2+	Bidir
6	MXC_N	Gigabit Ethernet MDI 2-	Bidir
7	MXD_P	Gigabit Ethernet MDI 1+	Bidir
8	MXD_N	Gigabit Ethernet MDI 1-	Bidir
YELA	VDD_3V3_SYS	Main 3.3V supply	Power
YELC	ETH_LED1	Yellow LED Cathode. On indicates activity.	Output
GRNA	VDD_3V3_SYS	Main 3.3V supply	Power
GRNC	ETH_LED0	Green LED Cathode. On for 1000Mbps link. Off for	Output
PSE Header - J13			
1	PSE_POS0	MCT	-
2	PSE_NEG0	MCT	-

Pin #	Schematic Signal	Usage/Description	Type/Dir
LAN 2 - J10-2			
1	MXA_P	Gigabit Ethernet MDI 4+	Bidir
2	MXA_N	Gigabit Ethernet MDI 4-	Bidir
3	MXB_P	Gigabit Ethernet MDI 3+	Bidir
4	MXB_N	Gigabit Ethernet MDI 3-	Bidir
5	MXC_P	Gigabit Ethernet MDI 2+	Bidir
6	MXC_N	Gigabit Ethernet MDI 2-	Bidir
7	MXD_P	Gigabit Ethernet MDI 1+	Bidir
8	MXD_N	Gigabit Ethernet MDI 1-	Bidir
YELA	VDD_3V3_SYS	Main 3.3V supply	Power
YELC	ETH_LED1	Yellow LED Cathode. On indicates activity.	Output
GRNA	VDD_3V3_SYS	Main 3.3V supply	Power
GRNC	ETH_LED0	Green LED Cathode. On for 1000Mbps link. Off for	Output
PSE Header - J20			
1	PSE_POS1	MCT	-
2	PSE_NEG1	MCT	-

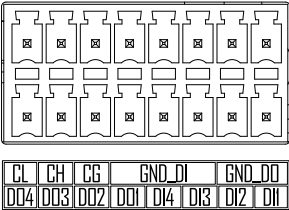
Pin #	Schematic Signal	Usage/Description	Type/Dir
LAN 3 - J10-3			
1	MXA_P	Gigabit Ethernet MDI 4+	Bidir
2	MXA_N	Gigabit Ethernet MDI 4-	Bidir
3	MXB_P	Gigabit Ethernet MDI 3+	Bidir
4	MXB_N	Gigabit Ethernet MDI 3-	Bidir
5	MXC_P	Gigabit Ethernet MDI 2+	Bidir
6	MXC_N	Gigabit Ethernet MDI 2-	Bidir
7	MXD_P	Gigabit Ethernet MDI 1+	Bidir
8	MXD_N	Gigabit Ethernet MDI 1-	Bidir
YELA	VDD_3V3_SYS	Main 3.3V supply	Power
YELC	ETH_LED1	Yellow LED Cathode. On indicates activity.	Output
GRNA	VDD_3V3_SYS	Main 3.3V supply	Power
GRNC	ETH_LED0	Green LED Cathode. On for 1000Mbps link. Off for 10/100Mbps.	Output
PSE Header - J23			
1	PSE_POS2	MCT	-
2	PSE_NEG2	MCT	-

Pin #	Schematic Signal	Usage/Description	Type/Dir
LAN 4 - J10-4			
1	MXA_P	Gigabit Ethernet MDI 4+	Bidir
2	MXA_N	Gigabit Ethernet MDI 4-	Bidir
3	MXB_P	Gigabit Ethernet MDI 3+	Bidir
4	MXB_N	Gigabit Ethernet MDI 3-	Bidir
5	MXC_P	Gigabit Ethernet MDI 2+	Bidir
6	MXC_N	Gigabit Ethernet MDI 2-	Bidir
7	MXD_P	Gigabit Ethernet MDI 1+	Bidir
8	MXD_N	Gigabit Ethernet MDI 1-	Bidir
YELA	VDD_3V3_SYS	Main 3.3V supply	Power
YELC	ETH_LED1	Yellow LED Cathode. On indicates activity.	Output
GRNA	VDD_3V3_SYS	Main 3.3V supply	Power
GRNC	ETH_LED0	Green LED Cathode. On for 1000Mbps link. Off for 10/100Mbps.	Output
PSE Header - J24			
1	PSE_POS3	MCT	-
2	PSE_NEG3	MCT	-
Note: 1.PSE 802.3 af, the power supply can be up to 15W. 2.With 4 ports simultaneously connected to PSE 802.3 af, the power supply capacity can reach 60W. 3.Gigabit Ethernet (10/100/1000M)			

Back View



DI/DO - J2



Type	Label Name	Schematic	Module Pin	BGA Number	GPIO	V/A Limits	Note
Input	DI1	DI_1_GPIO01	118	PQ.05	453	12V	12V Digital Input, ground signal needs to be connected to GND_DI (Pin2/4/6)
	DI2	DI_2_GPIO09	211	PAC.06	492		
	DI3	DI_3_GPIO11	216	PQ.06	454		
	DI4	DI_4_GPIO13	228	PH.00	391		
Output	DO1	DO_1_GPIO	193	PI.00	399	40V/500mA load per pin	Digital output, maximum withstand voltage 40V, ground signal needs to be connected to GND_DO(Pin8/10)
	DO2	DO_2_GPIO	195	PI.01	400		
	DO3	DO_3_GPIO	197	PI.02	401		
	DO4	DO_4_GPIO	199	PH.07	398		
Ground	GND_DI	/					The reference ground signal for the 12V Digital Input, which is also the return path for the DI
	GND_DO						The reference ground signal of the digital output, which is also the return path of the DO
	CG						The reference ground signal for CAN
CAN	CH	/					CAN bus with standard differential signals, ground signal needs to be connected to GND_ISO (Pin 12)
	CL						

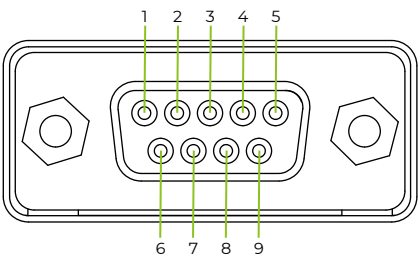
Note:

1. The Controller Area Network (CAN) enables communication among devices. CAN0_Low Differential CAN signal negative level; CAN0_High Differential CAN signal positive level

2. Support CAN FD (Controller Area Network Flexible Data-Rate) protocol at 5Mbps.

3. There is a Terminating Resistor Switch. CAN_I20R_EN=L to enable terminating resistor. This can be controlled by the GPIO, refer to the wiki for more information on how to operate it

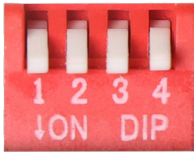
COM 232/422/485 - J1



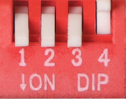
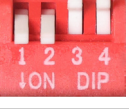
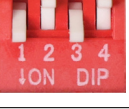
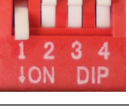
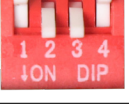
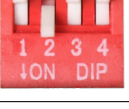
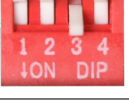
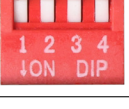
MODE	001	000/100	010/110
Pin #	RS232	RS422	RS485
1		TXD-	Data-
2	RXD	TXD+	Data+
3	TXD	RXD+	
4		RXD-	
5	GND	GND	GND
6			
7	RTS		
8	CTS		
9			

Note:
1.Default mode is RS485.
2.Toggle the switch(SW3) for mode switching after power off to avoid damage to the device.
3.RS232 comes with flow control.

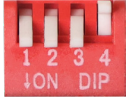
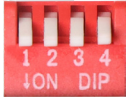
Mode selection instructions



Toggle the switch(SW3) for switching the mode of COM RS232/RS422/RS485 (DB9 J1).

	MODE_0	MODE_1	MODE_2	Mode	Status
	0	0	0	RS-422 Full Duplex	1T/1R RS-422
	0	0	1	Pure RS-232	3T/5R RS-232
	0	1	0	RS-485 Half Duplex	1T/1R RS-485 ,TX ENABLE Low Active
	0	1	1	RS-485 Half Duplex	1T/1R RS-485 ,TX ENABLE High Active
	1	0	0	RS-422 Full Duplex	1T/1R RS-422 with termination resistor
	1	0	1	Pure RS-232	1T/1R RS-232 co-exists with RS485 application without the need for the bus switch IC (for special usage).
	1	1	0	RS-485 Half Duplex	1T/1R RS-485 with termination resistor TX ENABLE Low Active
	1	1	1	Low Power Shutdown	All I/O pins are High Impedance

Slew rate control pin, Logical Low input will limit driver slew from either RS-232 to 1Mbps or RS-485 to 10Mbps. Internal pull high= 625KΩ.

	Status	Note
	OFF	SLEW = Vcc RS-232: Maximum Data Rate is 1.5Mbps RS-485/RS-422: Maximum Data Rate is 10Mbps The actual MaximumData Rate depends on the JetsonSOMused
	ON	SLEW = GND RS-232: Maximum Data Rate is 250Kbps RS-485/RS-422: Maximum Data Rate is 250kbps The actual MaximumData Rate depends on the JetsonSOMused

HDMI Port - J16

Pin #	Module Pin Name	Module Pin #	Schematic Signal	Usage/Description	Type/Dir
1	DP1_TXD0 (HDMI_TXD2)_P	65	HDMI_TXD2_CON_P	HDMI Transmit Data 2+	Output
2	–	–		Ground	Ground
3	DP1_TXD0 (HDMI_TXD2)_N	63	HDMI_TXD2_CON_N	HDMI Transmit Data 2–	Output
4	DP1_TXD1 (HDMI_TXD1)_P	71	HDMI_TXD1_CON_P	HDMI Transmit Data 1+	Output
5	–	–		Ground	Ground
6	DP1_TXD1 (HDMI_TXD1)_N	69	HDMI_TXD1_CON_N	HDMI Transmit Data 1–	Output
7	DP1_TXD2 (HDMI_TXD0)_P	77	HDMI_TXD0_CON_P	HDMI Transmit Data 0+	Output
8	–	–		Ground	Ground
9	DP1_TXD2 (HDMI_TXD0)_N	75	HDMI_TXD0_CON_N	HDMI Transmit Data 0–	Output
10	DP1_TXD3 (HDMI_TXC)_P	83	HDMI_TXC_CON_P	HDMI Transmit Clock+	Output
11	–	–		Ground	Ground
12	DP1_TXD3 (HDMI_TXC)_N	81	HDMI_TXC_CON_N	HDMI Transmit Clock–	Output
13	HDMI_CEC	94	HDMI_CEC	HDMI CEC	Bidir
14	–	–		Unused	Unused
15	DP1_AUX_P (HDMI_DDC_SCL)	100	HDMI_DDC_SCL_5V0	HDMI DDC Clock	Output /OD
16	DP1_AUX_N (HDMI_DDC_SDA)	98	HDMI_DDC_SDA_5V0	HDMI DDC Data	Bidir/OD
17	–	–		Ground	Ground
18	–	–	VDD_5V0_HDMI_CO	HDMI 5V Power	Power
19	DP1_HPD (HDMI_HPD)	96	HDMI_HPD_CON	HDMI Hot Plug Detect	Input

Note:
1. Support hot-swapped

USB3.1 TYPE-A x2 - J3

Pin #	Module Pin Name1	Module Pin #	Schematic Signal	Usage/Description	Type/Dir2
USB 3.1 Type A					
1	–	–	–	VBUS Supply	Power
2	USB1_D_N	115	USB2_A_N	USB 2.0 #3 Data from hub	Bidir
3	USB1_D_P	117	USB2_A_P		
4	–	–		Ground	Ground
5	USBSS_RX_N	161	USBSS_RX_A_CON2_N	USB 3.1 Receive Data from hub	Input
6	USBSS_RX_P	163	USBSS_RX_A_CON2_P		
7	–	–		Ground	Ground
8	USBSS_TX_N	166	USBSS_TX_A_CON2_N	USB 3.1 Transmit Data from hub	Output
9	USBSS_TX_P	168	USBSS_TX_A_CON2_P		
USB 3.1 Type A					
1	–	–	–	VBUS Supply	Power
2	USB1_D_N	115	USB1_A_N	USB 2.0 #3 Data from hub	Bidir
3	USB1_D_P	117	USB1_A_P		
4	–	–		Ground	Ground
5	USBSS_RX_N	161	USBSS_RX_A_CON1_N	USB 3.1 Receive Data from hub	Input
6	USBSS_RX_P	163	USBSS_RX_A_CON1_P		
7	–	–		Ground	Ground
8	USBSS_TX_N	166	USBSS_TX_A_CON1_N	USB 3.1 Transmit Data from hub	Output
9	USBSS_TX_P	168	USBSS_TX_A_CON1_P		
Note: 1. The upper and lower USB ports share a current-limiting IC, with a total power supply capacity of 2.1A maximum output current (single can also be 2.1A). If over 2.1A, it will enter the over-current protection state. 2. The module pin names not directly connected to the USB connector pins but are routed to the input of the USB hub. 3. Orin NX module comes with 3 USB3.2, only one of which is used in reComputer and converted to 4 ways via USB3.1 hub. (USB3.1 TYPE-A x2 - J3 and USB3.1 TYPE-A x2 -J4). 4. The Type A is 5Gbps for all two, and only supports USB Host, not Device mode. 5. Provide 5V 2.1A. 6. Hot-swappable.					

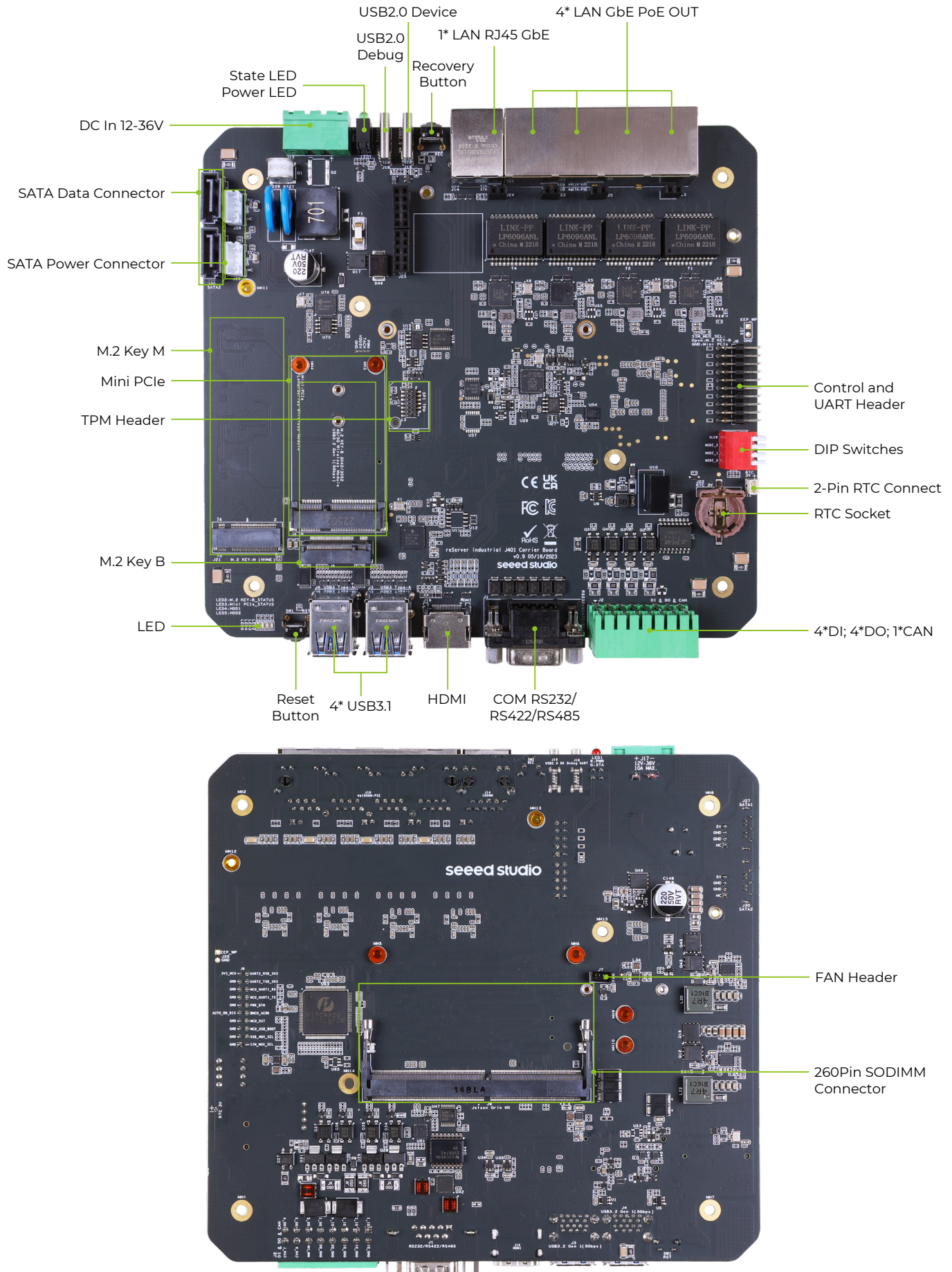
USB3.1 TYPE-A x2 - J4

Pin #	Module Pin Name1	Module Pin #	Schematic Signal	Usage/Description	Type/Dir2
USB 3.1 Type A					
1	–	–	–	VBUS Supply	Power
2	USB1_D_N	115	USB2_A_N	USB 2.0 #3 Data from hub	Bidir
3	USB1_D_P	117	USB2_A_P		
4	–	–		Ground	Ground
5	USBSS_RX_N	161	USBSS_RX_A_CON2_N	USB 3.1 Receive Data from hub	Input
6	USBSS_RX_P	163	USBSS_RX_A_CON2_P		
7	–	–		Ground	Ground
8	USBSS_TX_N	166	USBSS_TX_A_CON2_N	USB 3.1 Transmit Data from hub	Output
9	USBSS_TX_P	168	USBSS_TX_A_CON2_P		
USB 3.1 Type A					
1	–	–	–	VBUS Supply	Power
2	USB1_D_N	115	USB1_A_N	USB 2.0 #3 Data from hub	Bidir
3	USB1_D_P	117	USB1_A_P		
4	–	–		Ground	Ground
5	USBSS_RX_N	161	USBSS_RX_A_CON1_N	USB 3.1 Receive Data from hub	Input
6	USBSS_RX_P	163	USBSS_RX_A_CON1_P		
7	–	–		Ground	Ground
8	USBSS_TX_N	166	USBSS_TX_A_CON1_N	USB 3.1 Transmit Data from hub	Output
9	USBSS_TX_P	168	USBSS_TX_A_CON1_P		
Note: 1. The upper and lower USB ports share a current-limiting IC, with a total power supply capacity of 2.1A maximum output current (single can also be 2.1A). If over 2.1A, it will enter the over-current protection state. 2. The module pin names not directly connected to the USB connector pins but are routed to the input of the USB hub. 3. Orin NX module comes with 3 USB3.2, only one of which is used in reComputer and converted to 4 ways via USB hub. (USB3.1 TYPE-A x2 - J3 and USB3.1 TYPE-A x2 -J4). 4. The Type A is 5Gbps for all two, and only supports USB Host, not Device mode. 5. Provide 5V 2.1A. 6. Hot-swappable.					

RESET Button

Reset button: the reset button executes a hard reset on the system.

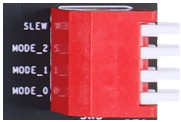
Carrier board



Control and UART Header - J8

PIN	Net Name	PIN	Net Name
1	GND	2	SIM_MUX_SEL
3	GND	4	USB_MUX_SEL
5	GND	6	MCU_USB_BOOT
7	GND	8	MCU_RST
9	AUTO_ON_DIS	10	BMCU_ACOK
11	GND	12	PWR_BTN*
13	GND	14	MCU_UART1_TX
15	GND	16	MCU_UART1_RX
17	GND	18	UART2_TXD_3V3
19	3V3_MCU	20	UART2_RXD_3V3

DIP Switches - SW3



Toggle the switch(SW3) for switching the mode of COM RS232/RS422/RS485 (DB9 J1).

RTC

RTC-Pin Header - J12

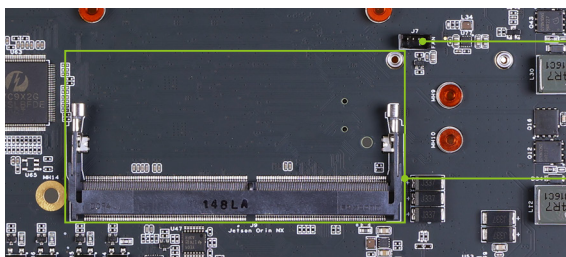
Pin #	Module Pin Name	Usage/Description	Type/Dir Default
1	–	Ground	Ground
2	PMIC_BBAT	RTC Back-up battery power	Power

Note:
Support 3V Coin Cell Battery ,2-pin 1.25MM

RTC-Coin Cell Battery Socket - J22

Pin #	Module Pin Name	Usage/Description	Type/Dir Default
1	–	Ground	Ground
2	PMIC_BBAT	RTC Back-up battery power	Power

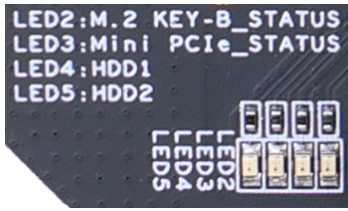
Note:
Support 3V Coin Cell Battery , CR1220



FAN Header

260Pin SODIMM
Connector

LED



LED2: M.2 Key B

LED3: Mini PCIe

LED4: HDD1

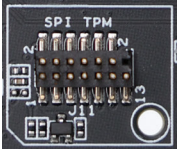
LED5: HDD2

M.2 Key M - J21

Pin #	Module Pin Name	Module Pin #	Usage/Description	Type/Dir Default	Pin #	Module Pin Name		Usage/Description	Type/Dir Default
1	–	–	Ground	Ground	2	–	–	Main 3.3V Supply	Power
3	–	–	Ground	Ground	4	–	–	–	–
5	PCIE0_RX3_N	155	PCle IF #0 Lane 3 Receive	Input	6	–	–	Unused	Unused
7	PCIE0_RX3_P	157	–	–	8	–	–	–	–
9	–	–	Ground	Ground	10	–	–	–	–
11	PCIE0_TX3_N	154	PCle IF #0 Lane 3 Transmit	Output	12	–	–	Main 3.3V Supply	Power
13	PCIE0_TX3_P	156	–	–	14	–	–	–	–
15	–	–	Ground	Ground	16	–	–	–	–
17	PCIE0_RX2_N	149	PCle IF #0 Lane 2 Receive	Input	18	–	–	–	–
19	PCIE0_RX2_P	151	–	–	20	–	–	–	–
21	–	–	Ground	Ground	22	–	–	–	–
23	PCIE0_TX2_N	148	PCle IF #0 Lane 2 Transmit	Output	24	–	–	–	–
25	PCIE0_TX2_P	150	–	–	26	–	–	–	–
27	–	–	Ground	Ground	28	–	–	–	–
29	PCIE0_RX1_N	137	PCle IF #0 Lane 1 Receive	Input	30	–	–	–	–
31	PCIE0_RX1_P	139	–	–	32	–	–	–	–
33	–	–	Ground	Ground	34	–	–	–	–
35	PCIE0_TX1_N	140	PCle IF #0 Lane 1 Transmit	Output	36	–	–	–	–
37	PCIE0_TX1_P	142	–	–	38	–	–	–	–
39	–	–	Ground	Ground	40	–	–	–	–
41	PCIE0_RX0_N	131	PCle IF #0 Lane 0 Receive	Input	42	–	–	–	–
43	PCIE0_RX0_P	133	–	–	44	GPIO06	130	M.2 Key M Alert	Input, 1.8V
45	–	–	Ground	Ground	46	–	–	–	–
47	PCIE0_TX0_N	134	PCle IF #0 Lane 0 Transmit	Output	48	–	–	–	–
49	PCIE0_TX0_P	136	–	–	50	PCIE0_RST	181	PCle IF #0 Reset	Output, 3.3V
51	–	–	Ground	Ground	52	PCIE0_CLKRE	180	PCle IF #0 Clock Request	Input, 3.3V
53	PCIE0_CLK_N	160	PCle IF #0 Reference Clock	Output	54	PCIE0_WAKE	179	PCle Wake (Level Shifted from 3.3V to 1.8V)	Input, 3.3V
55	PCIE0_CLK_P	162	–	–	56	–	–	–	–
57	–	–	Ground	Ground	58	–	–	–	–
59	–	–	–	–	60	–	–	–	–
61	–	–	Unused (Key)	Unused	62	–	–	–	–
63	–	–	–	–	64	–	–	–	–
65	–	–	–	–	66	–	–	–	–
67	–	–	–	–	68	–	–	32KHz Suspend Clock	Output, 3.3V
69	–	–	–	–	70	–	–	–	–
71	–	–	–	–	72	–	–	Main 3.3V Supply	Power
73	–	–	Ground	Ground	74	–	–	–	–
75	–	–	–	–	–	–	–	–	–

Note:
Support PCIe Gen4.0 SSD

TPM Header - J11



PIN	Schematic Signal	PIN	Schematic Signal
1	VDD_TPM	2	SPIO_IRQ_LS
3	SPIO_RST_LS	4	NC
5	NC	6	NC
7	VDD_TPM	8	GND
9	SPIO_CS0_LS	10	SPIO_SCK_LS
11	SPIO_MISO_LS	12	PIO_MOSI_LS
13	NC	14	NC

Note:
Support SPI interface TPM module, test with TPM2.0 Module with infineon SLB9670 (module not included)

Mini PCIe - J5

Pin #	Module Pin Name	Module Pin #	Schematic Signal/Description	Type/Dir Default	Pin #	Module Pin Name	Module Pin #	Schematic Signal/Description	Type/Dir Default
1	-	-	PCIe_WAKE_3V3 #I2C Expansion IO-GPIO304	-	2	-	-	VDD_3V3_SYS	Power
3	-	-	NC	-	4	-	-	GND	Ground
5	-	-	NC	-	6	-	-	NC	-
7	-	-	NC	-	8	-	-	PCIe_USIM_PWR	-
9	-	-	GND	Ground	10	-	-	PCIe_USIM_DATA	-
11	UART0_TXD	99	UART0_TXD_3V3	Output	12	-	-	PCIe_USIM_CLK	-
13	UART0_RXD	101	UART0_RXD_3V3	Input	14	-	-	PCIe_USIM_RST	-
15	-	-	GND	Ground	16	-	-	NC	-
17	-	-	NC	-	18	-	-	GND	Ground
19	-	-	-	-	20	GPIO05	128	W_DISABLE1#_3V3	Output
21	-	-	GND	Ground	22	-	-	M2B_PCIe_Reset #I2C Expansion IO-GPIO309	-
23	UART0_RTS	103	UART0_RTS_3V3	Output	24	-	-	VDD_3V3_SYS	Power
25	UART0_CTS	105	UART0_CTS_3V3	Input	26	-	-	GND	Ground
27	-	-	GND	Ground	28	-	-	NC	-
29	-	-	GND	Ground	30	I2C1_SCL	189	I2C1_SCL	Output
31	-	-	-	-	32	I2C1_SDA	191	I2C1_SDA	Bidir
33	-	-	NC	-	34	-	-	GND	Ground
35	-	-	GND	Ground	36	USB2_D_N	121	M2B_PCIe_HSD4_N	Bidir
37	-	-	GND	Ground	38	USB2_D_P	123	M2B_PCIe_HSD4_P	Bidir
39	-	-	VDD_3V3_SYS	Power	40	-	-	GND	Ground
41	-	-	VDD_3V3_SYS	Power	42	-	-	LED_WWAN	-
43	-	-	GND	Ground	44	-	-	NC	-
45	SPI1_SCK	106	SPI1_SCK_3V3	Output	46	-	-	NC	-
47	SPI1_MISO	108	SPI1_MISO_3V3	Input	48	-	-	NC	-
49	SPI1_MOSI	104	SPI1_MOSI_3V3	Output	50	-	-	GND	Ground
51	SPI1_CS0	110	SPI1_CS0_3V3	Input	52	-	-	VDD_3V3_SYS	Power

Note:
1.Support 4G/LoRaWAN®/GPS.
2.When using Mini PCIe 4G module, you need to set SIM_MUX_SEL to low level (Pin1 Pin2 jumper cap of J8 is connected), when using LoRa module, you need to disconnect SIM card signal, at this time you need to set SIM_MUX_SEL to high level(Pin1 Pin2 jumper cap of J8 is disconnected).
3.Can be used as UART, IIC, GPIO.
4.M.2 Key B(J6) and Mini PCIe(J5) cannot be used at the same time.

M.2 Key B - J6

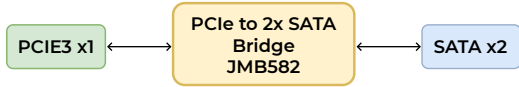
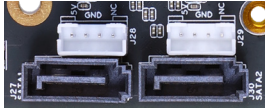
Pin #	Module Pin Name	Module Pin #	Schematic Signal/Description	Type/Dir Default	Pin #	Module Pin Name	Module Pin #	Schematic Signal/Description	Type/Dir Default
1	-	-	NC	-	2	-	-	VDD_3V3_SYS	Power
3	-	-	GND	Ground	4	-	-	VDD_3V3_SYS	Power
5	-	-	GND	Ground	6	GPIO10	212	M2B_POWER_OFF	Output
7	USB2_D_P	123	M2B_PCl_e_HSD4_P	Bidir	8	GPIO05	128	W_DISABLE1#_3V3	Output
9	USB2_D_N	121	M2B_PCl_e_HSD4_N	Bidir	10	-	-	M2B_STATUS_LED	-
11	-	-	GND	Ground	12	-	-	-	-
13	-	-	-	-	14	-	-	-	-
15	-	-	-	-	16	-	-	-	-
17	-	-	-	-	18	-	-	-	-
19	-	-	-	-	20	I2S1_SCLK	226	I2S1_SCLK	Output
21	-	-	NC	-	22	I2S1_DOUT	220	I2S1_SDOUT	Output
23	-	-	M2B_WOWWAN#_1V8	-	24	I2S1_DIN	222	I2S1_SDIN	Output
25	-	-	M2B_DPR_1V8	-	26	GPIO03	126	W_DISABLE2#	Output
27	-	-	GND	Ground	28	I2S1_FS	224	I2S1_LRCK	Input
29	DP0_TXD2_N/USBSS2_RX_N	51	M2B_SSRX4_N	Input	30	-	-	M2B_USIM_RST	-
31	DP0_TXD2_P/USBSS2_RX_P	53	M2B_SSRX4_P	Input	32	-	-	M2B_USIM_CLK	-
33	-	-	GND	Ground	34	-	-	M2B_USIM_DATA	-
35	DP0_TXD3_N/USBSS2_TX_N	57	M2B_SSTX4_N	Output	36	-	-	M2B_USIM_PWR	-
37	DP0_TXD3_P/USBSS2_TX_P	59	M2B_SSTX4_P	Output	38	SPI1_CS1	112	M.2_WLAN_Tx_EN_1V8	Output
39	-	-	GND	Ground	40	-	-	NC	-
41	-	-	NC	-	42	-	-	NC	-
43	-	-	NC	-	44	-	-	NC	-
45	-	-	GND	Ground	46	-	-	NC	-
47	-	-	NC	-	48	-	-	NC	-
49	-	-	NC	-	50	-	-	NC	-
51	-	-	GND	Ground	52	-	-	NC	-
53	-	-	NC	-	54	-	-	NC	-
55	-	-	NC	-	56	-	-	NC	-
57	-	-	GND	Ground	58	-	-	NC	-
59	-	-	-	-	60	SPI0_CS1	97	LAA_n79_Tx_EN_1V8	Output
61	-	-	-	-	62	UART0_TXD	99	UART0_TXD	Output
63	-	-	-	-	64	UART0_RXD	101	UART0_RXD	Input
65	-	-	-	-	66	-	-	SIM_DETECT_1V8	-
67	-	-	Device_Reset	-	68	CLK_32K_OUT	210	M2B_SUSCLK_32KHZ	Output
69	-	-	NC	-	70	-	-	VDD_3V3_SYS	Power
71	-	-	GND	Ground	72	-	-	VDD_3V3_SYS	Power
73	-	-	GND	Ground	74	-	-	VDD_3V3_SYS	Power
75	-	-	NC	-	76	-	-	GND	Ground
77	-	-	GND	Ground					

Note:

1. Support 4G、5G
2. When using M.2 module, you need to set SIM_MUX_SEL to high level, i.e. Pin1 Pin2 jumper cap of J8 is disconnected.
3. Support UART, IIS, GPIO
4. LED2 is the M.2 status indicator, indicating whether M.2 is turned on and off normally, M2B_STATUS_LED
5. M.2 Key B(J6) and Mini PCIe(J5) cannot be used at the same time.
6. M.2 Key B 3042/3052

SATA - J27, J30

SATA Data Connector



Pin #	Schematic Signal	Usage/Description	Type/Dir2
SATA PORT 0			
G1	–	Ground	Ground
T+	S_TXP0		Input
T-	S_TXN0		
G2		Ground	Ground
R-	S_RXN0		Output
R+	S_RXP0		
G3		Ground	Ground
S1	–	Ground	Ground
S2	–	Ground	Ground
SATA PORT 1			
G1	–	Ground	Ground
T+	S_TXP1		Input
T-	S_TXN1		
G2		Ground	Ground
R-	S_RXN1		Output
R+	S_RXP1		
G3		Ground	Ground
S1	–	Ground	Ground
S2	–	Ground	Ground
Note: 1. Support SATA III 6.0Gbps. 2. Support 2.5" SATA SSDs, SATA HDDs, and 4T HDDs up to 15mm thick.			

SATA Power Connector - J28, 29

Pin #	Schematic Signal
SATA Power Connector - J28	
1	NC
2	GND
3	GND
4	SATA_5V
SATA Power Connector - 29	
1	NC
2	GND
3	GND
4	SATA_5V

FAN Header -J7

Pin #	Module Pin Name	Module Pin #	Schematic Signal
1	–	–	
2	–	–	
3	GPIO08 (SDMMC_CD)	208	FAN_TACH
4	GPIO14 (PWM)	230	FAN_PWM

Note:
 1. Whether the fan is self-starting or starts only when it reaches the temperature can be set by yourself.
 2. Connect 4Pin 5V fan, fan speed can be controlled, and the speed can be detected.

Accessories Information

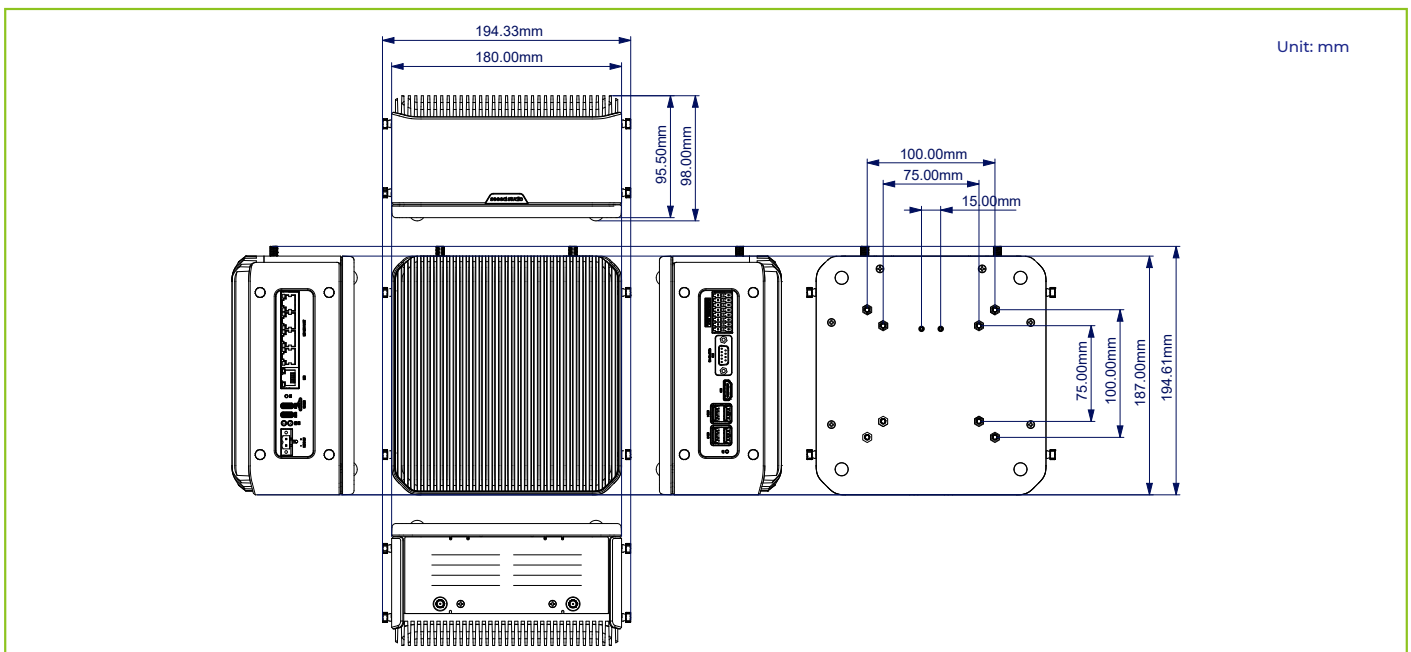
Please kindly know that the accessories below are not included in the package. We provide the information here to help you choose the appropriate accessories.

Type	Product Name
GPS	EC25EUXGA-MINIPICIE;
4G	EC25EUXGA-MINIPICIE; EC20CEHCLG-MINIPICIE
5G	SIM8202G-M2
LoRaWAN	Wio-WM1302 LoRaWAN Gateway Module (SPI) - US915 - M
TPM	TPM2.0 module with Infineon SLB9670

Pre-installed OS

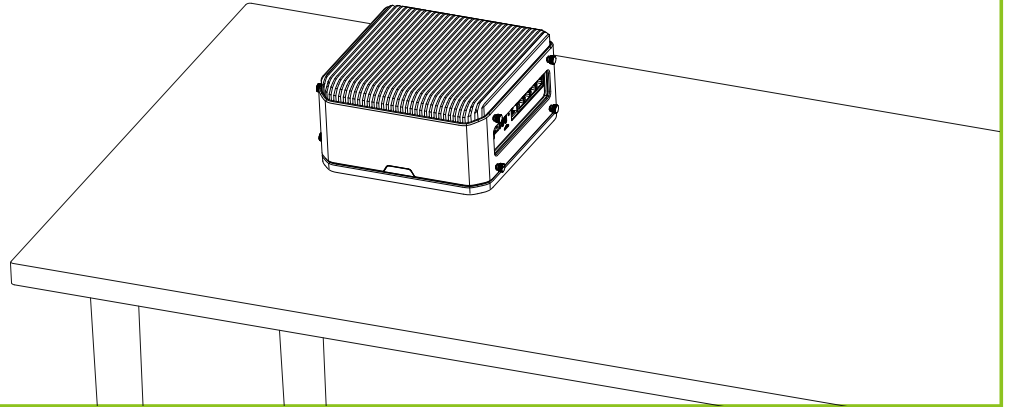
Pre-installed Jetpack 5.1.1 (above) (provide Linux OS with board support package). reServer Industrial comes with M.2 NVMe 2280 SSD 128G with OS pre-installed in the SSD. If you need to reflash the image, please refer to our wiki for details.

Dimensions

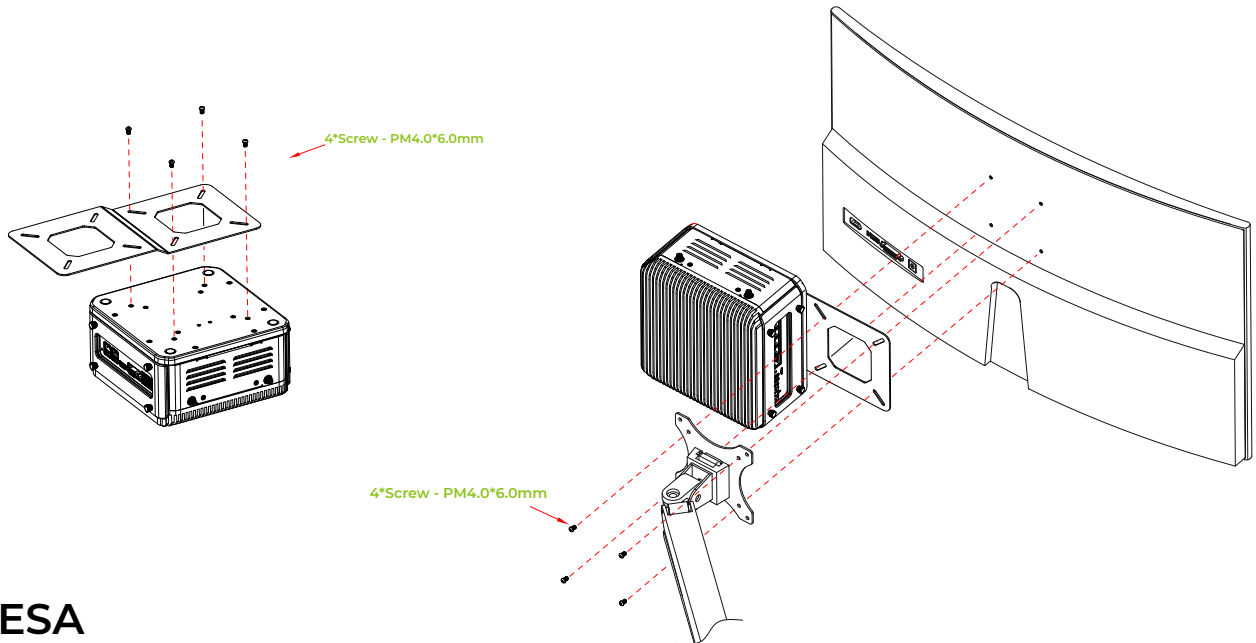


Flexible mounting

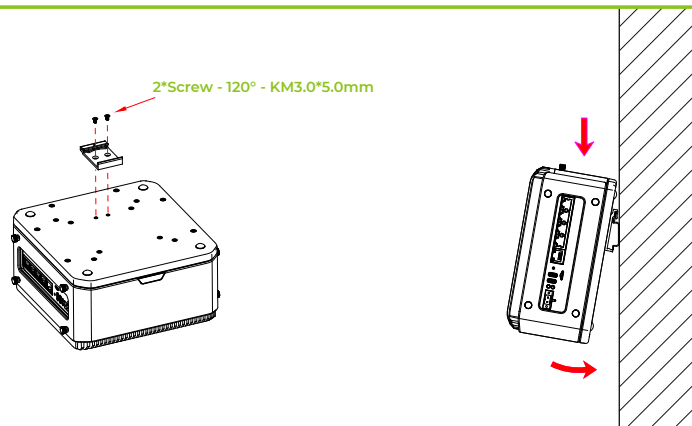
Desk



VESA



DIN rail



Certifications



More information

Please check our wiki to learn more about this device and if you have any questions, feel free to reach out to our Forum and Discord community.



Scan for more information



Wiki



Forum



Discord

For more information, you can also refer to NVIDIA official Jetson Download Center